Ali EL KAOUTAR

Engineering student at ENSEEIHT — Computer Science

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PROFILE

I'm a first-year engineering student at ENSEEIHT in Toulouse, where I'm studying computer science. After earning my baccalauréat, I spent two years in classes préparatoires in Morocco. That path gave me a strong foundation in mathematics and physics, and taught me to work rigorously and autonomously under pressure.

I'm particularly interested in **artificial intelligence and cybersecurity**, two fields that increasingly shape each other. AI is changing how systems are defended and attacked, while securing those systems has become essential as they spread into everyday life.

EDUCATION

ENSEEIHT — Toulouse INP (Institut National Polytechnique) — Toulouse, France
Engineering degree, 1st year — Digital Sciences department (Computer Science) — *Sep. 2026 – Present*

CPGE — Classes Préparatoires aux Grandes Écoles — Agadir, Morocco
CPGE Reda Slaoui — two-year intensive programme in mathematics and physics — *Sep. 2024 – Jun. 2026*

Baccalauréat — with highest honours (mention Très bien) — Agadir, Morocco
Al Qalam High School — *Jun. 2024*

ACADEMIC PROJECTS

TIPE — The Travelling Salesman Problem — CPGE
Algorithms, complexity and optimisation heuristics — *2024 – 2026*

- Studied the problem and exact solving methods, and showed their practical limit due to the problem's intrinsic factorial complexity.
- Implemented and benchmarked several heuristics to quickly obtain good approximate solutions, and analysed their performance.
- Applied the algorithms to a real-world case: optimising a route connecting cities in Morocco.

SKILLS & INTERESTS

Programming: Python, C, Ada, MATLAB
Languages: Arabic (native), French (fluent), English (fluent)
Interests: Rock climbing